

Announcements – Jan 22

- HW 1 is due Friday at 9am.
- TA offices hours are posted. Office hours start this week.
- SI sessions are Tue 6-7pm and Thu 2-3pm.
- Quiz 3 trivia – The Apple II could display 6 colors in hi-res. It used a very convoluted system so that certain combination of colors could not be displayed. 11% of you got the correct answer. The most common answer was "16" which in many ways made the most sense.

Two Categories of Propositions

Definition: Tautology

A proposition that always evaluates to true.

examples: T , $p \vee \neg p$

Definition: Contradiction

A proposition that always evaluates to false.

examples: F , $p \wedge \neg p$

For example, show whether

$$p \oplus (q \wedge r)$$

is a tautology or a contradiction or neither?

$$F \oplus (F \wedge F) = F \oplus F = F$$

so not a tautology

$$T \oplus (F \wedge F) = T \oplus F = T$$

so not a contradiction

Satisfiable Proposition

A proposition is ***Satisfiable*** if there is an assignment of truth values to its variables that makes it true.

In other words: a proposition is satisfiable iff it is not a contradiction.

Example: Show $(p \wedge q) \rightarrow \neg p$ is satisfiable

we must find an assignment of values for p and q that make the statement true

This is easy since all we need to do is make the hypothesis false

p = true and q = false will work.

$$t \wedge f \rightarrow \neg t \equiv f \rightarrow f \equiv t$$

Biconditionals and Logical Equivalence

Definition: Logically Equivalent

Compound propositions p and q are logically equivalent ($p \equiv q$) if $p \leftrightarrow q$ is a tautology.

In other words, the truth tables for p and q are the same.

Laws of Propositional Logic

We can use known logical equivalences to simplify propositions.
Here are a few of the named equivalences:

<i>Equivalence</i>	<i>Name</i>
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

Laws of Propositional Logic

$$p \rightarrow q \equiv \neg p \vee q$$

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

$$p \vee q \equiv \neg p \rightarrow q$$

$$p \wedge q \equiv \neg(p \rightarrow \neg q)$$

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

$$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$$

$$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$$

$$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$$

$$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

Applications of Logical Equivalences

Question: Is $(p \wedge q) \rightarrow p$ a tautology? (1)

By use of a Truth Table;

p	q	$p \wedge q$	$(p \wedge q) \rightarrow p$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

Because the expression evaluates to true for all possible arrangements of truth values, the expression is a tautology.

Applications of Logical Equivalences

Question: Is $(p \wedge q) \rightarrow p$ a tautology? (2)

By application of logical equivalences:

$$(p \wedge q) \rightarrow p \equiv \neg (p \wedge q) \vee p$$

$$\equiv \neg p \vee \neg q \vee p$$

$$\equiv \neg p \vee p \vee \neg q$$

$$\equiv p \vee \neg p \vee \neg q$$

$$\equiv T \vee \neg q$$

$$\equiv \neg q \vee T$$

$$\equiv T$$

- Conditional Identities

-DeMorgan's Laws

-Commutative Laws

-Commutative Laws

-Complement Laws

-Commutative Laws

-Domination Laws

This is correct, but I personally think this is more steps than we need.

Applications of Logical Equivalences

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By application of logical equivalences:

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$$\equiv \neg p \vee \neg q \vee p$$

$$\equiv \neg p \vee p \vee \neg q$$

$$\equiv T \vee \neg q$$

$$\equiv T$$

- Conditional Identities

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By Reasoning:

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When p is true:

$(T \wedge q) \rightarrow T$ is true regardless of q

(Why? By $\rightarrow$'s truth table, $? \rightarrow T \equiv T$!)

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When p is false:

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(Why? $F \wedge q \equiv F$, and $F \rightarrow F \equiv T$!)

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Thus, $(p \wedge q) \rightarrow p$ is a tautology

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What we just learned

Three quick ways to prove that something is a **tautology**:

- Truth Table
 - Do all cases resolve to TRUE ?
 - Logical Equivalences
 - Can we convert the expression to TRUE ?
 - Reasoning
 - Any argument you make
 - Our example did “proof by cases”
-

Proving that something is a contradiction?

How to prove that something is a **contradiction**?

- Truth Table
 - Do all cases resolve to FALSE ?
- Logical Equivalences
 - Can we convert the expression to FALSE ?
- Reasoning
 - Any argument you make

Applications of Logical Equivalences

Example(s): [Assume `games` is an integer value.]

```
if ( (games <= 10 or ties > 2) and games >= 11 ) :  
Let g:games <= 10           t:ties > 2           ¬g: games >= 11
```

Applications of Logical Equivalences

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$(g \vee t) \wedge \neg g \equiv (g \wedge \neg g) \vee (t \wedge \neg g)$ [Distribution]

$\equiv \mathbf{F} \vee (t \wedge \neg g)$ [Negation]

$\equiv t \wedge \neg g$ [Identity]

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$(g \vee t) \wedge \neg g \equiv (g \wedge \neg g) \vee (t \wedge \neg g)$ [Distribution]

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$\equiv t \wedge \neg g$ [Identity]

Thus, the statement can be rewritten more efficiently as:

```
if (ties > 2 and games >= 11) :
```

Satisfiability

- The book has a couple of examples of applications (Placing Queens, Sudoku) you may want to look at.
- How hard is it to tell if logical expression is satisfiable?
- If we have n variables, how many rows does the truth table have?
- 2^n
- This is called exponential growth . . . and it is bad.

Satisfiability

- If we had 100 variables, and could fill in 1,000 rows a second, it would take ~ 42 quintillion years to finish. That's 3 billion times longer than the universe has existed. (they think)
- Is there a smarter (faster) way to determine if a general Boolean expression is satisfiable?
- We don't think so.
- The satisfiability problem has historic importance since it was the first problem proved to be "NP-Complete". Which roughly means it is at least as hard as any of a wide class of problems.

End of Section 1.3
Starting Section 1.4

Predicates and Quantifiers

Definition: Predicate (a.k.a. Propositional Function)

A statement that includes at least one variable and will evaluate to either true or false when the variable(s) are assigned values.

Examples:

$P(x): x^2 = 4$ $P(-2)$ is true $P(3)$ is false

$Q(x): x$ is a student in 144

$R(x, y, z): x$ is the parent of y and z

Predicates and Quantifiers

The domain of a variable in a predicate is the set of all possible values for the variable.

Examples:

$P(x): x^2 = 4$ domain might be all integers

or all positive integers

or all real numbers

or even $\{0, 5, 7\}$

How do we know what the domain is?

We must be told.

Quantification

Idea: Establish truth of predicates over sets of values.

Two common generalizations:

(1) Universal Quantification ($\forall$)

– Reads “for every” or “for all”

(2) Existential Quantification ($\exists$)

– Reads “There exists”

$\exists!$ is another quantification

– Reads “There exists a unique”

$\forall x \ P(x)$ is true only when $P(x)$ is true for each x in the domain. (False otherwise.)

(easy to prove false, hard to prove true)

$\exists x \ P(x)$ is true when at least one domain member causes $P(x)$ to evaluate to true.

(easy to prove true, hard to prove false)